



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester M.C.A. (CBCS) Degree Examinations
 January 2024

22MCA353 – INDUSTRIAL AND MEDICAL IOT

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Define Sensor and mention the categories of sensor.	06	L*1	1	1
	b)	"IoT System is dynamic and Self configuring"- Justify your statement with example	04	L3	1	2
	c)	Differentiate between M2M and IoT.	10	L2	1	1
2.	a)	Compare the working of Push pull communication model with Exclusive pair communication model.	14	L2	1	1
	b)	Differentiate between Ipv4 and Ipv6.	06	L2	1	1
		Unit – II				
3.	a)	Compare REST with WebSocket Service.	10	L2	2	1
	b)	Explain the entities, objects and concepts defined in domain model.	10	L2	2	1
	a)	Explain the working of IoT level-4 with the block diagram.	10	L1	2	1
	b)	How can the functional group be mapped to operational view pecification in home automation. Explain.	10	L2	2	1
		Unit – III				
5.	a)	Write a python program for controlling an LED with a switch using Raspberry pi. Give the circuit setup.	10	L2	3	2
	b)	What is Arduino? Explain the program structure of Arduino.	10	L1	3	1
6.	a)	What is WAMP protocol? Discuss the key concepts of WAMP.	08	L2	3	1
	b)	Write Arduino program for LED fade in and fade out. Write steps for circuit setup.	12	L1	3	1
		Unit – IV				
7.	a)	Discuss an analysis framework for reviewing existing IoT taxonomies and their relevance to IIoT.	10	L1	4	1
	b)	Select 'Integration of Sensors for Equipment Monitoring' case study and present a detailed analysis of the implementation, outcomes, and lessons learned.	10	L2	4	1
	a)	Explain the difference between IoT and IIoT. Provide examples of applications for each.	10	L2	4	1
	b)	Examine the connectivity options used in IIoT deployments and their advantages and limitations.	10	L1	4	1
		Unit – V				
9.	a)	Explain the concept of IoMT and its relationship with Telehealth and Telemedicine. How do these technologies work together to improve healthcare services?	10	L1	5	1
	b)	Describe the advantages that IoMT offers to healthcare providers and institutions. How does IoMT enhance healthcare delivery and management from their perspective?	10	L2	5	1
10.	a)	Compare IoMT with IoT. Highlight the key differences and similarities between these two concepts, emphasizing their respective applications and requirements.	10	L2	5	1
	b)	Provide an overview of the various IoMT devices commonly used in healthcare. Explain their functions and how they contribute to patient care and data collection.	10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
